Exact Input-Side Minimization and 56-Addition Schemes on Two New Classes for Rank-23 3×3 Matrix Multiplication

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Research note — logic repo (https://github.com/gsidebottom/logic), matmul track. v1 2026-07-03 ("Three New 60-Addition, Rank-23 Schemes…", written against the Aug-2025 frontier of 60; full text in git history). v2 2026-07-04, revised after the record chain $60 \to 59 \to 58 \to 56$ and after our exact side-minimizer tied the 56 record with independent machinery — twice more, on two further de Groote classes. v3 2026-07-05 adds the database-wide census (§7). Companion to `doc/matmul_53_3x3_schemes.md`; every claim has a mechanical check (§10).

Abstract

The additive-complexity record for exact non-commutative 3×3 matrix multiplication with 23 multiplications, without change of basis, is **56 additions** (Yinqi Sun, arXiv:2604.27645, Apr 2026), reached through a rapid chain: 60 (Stapleton, Aug 2025) $\to$ 59 (Mårtensson–Stankovski Wagner–Stapleton, MWS) $\to$ 58 $\times 3$ (Perminov) $\to$ 56. We verified every scheme in that chain exactly (Brent over $\mathbb{Z}$, ternary coefficients, matching addition counts) and classified each against the Heule–Kauers–Seidl (HKS) database: **all five are HKS database classes**, and Sun's 56 is a cyclic-symmetry image of one of Perminov's 58-classes — the last two additions of the record came from a *change of representative*, not a new scheme.

Sun's count has two structural ingredients: input sides in *addition-chain form* (all 23 left/right factors are values of one 13-step chain), and per-input-side optimality certificates. Pair-extraction CSE (common-subexpression elimination: repeatedly compute the most frequent signed pair into a temporary and substitute it) — the engine behind $62 \to 60 \to 59 \to 58$ — cannot represent that structure; it plateaus at $58 = 14{+}14{+}30$ on Sun's own scheme. We therefore built the missing tool and report:

1. **An exact input-side minimizer** (`matmul/sidemin.py`): the minimum number of $\pm$-additions such that one chain over the 9 input cells contains every distinct multi-term factor row (up to sign), solved exactly by iterative deepening over *helper values* with a closure normal form (complete by an exchange argument; helpers may be arbitrary integer vectors, strictly more general than Sun's certificate model). It reproduces both of Sun's input-side optima (13, 13) *and* both of his impossibility certificates in 0.3 s, and its chains are replay-verified.

2. **The 56 record tied, three times over.** With exact input sides + a restart-greedy output side: Sun's representative gives **$56 = 13{+}13{+}30$ on all 24 enumerated sign models**. Perminov's scheme cn122 — class `i19w225c4efh`, published at 58 — gives **$56 = 13{+}14{+}29$ directly on his published representative**: the exact side-minimizer alone recovers two additions. And an orbit walk in the class of his cn119 — `i12w219c23ci`, also published at 58 — reaches a representative scoring **$56 = 13{+}14{+}29$** (verified at high effort). All three classes are pairwise inequivalent (exact refutations), so **56 additions is now attained on three inequivalent rank-23 classes** where before there was one.

3. **Every published count in the chain drops or ties on its own representative** under exact input sides: MWS's 59-scheme $\to$ 58 ($14{+}15{+}29$); Perminov's cn119 $\to$ 57, cn120 $\to$ 57 ($13{+}16{+}28$; this is Sun's class, so $C = 28$ exists *inside* the record class), cn122 $\to$ 56; and the three 60-addition classes of this note's v1 $\to$ **59 each** ($15{+}15{+}29$). Stapleton's class alone stays at 60 ($16{+}16{+}28$).

4. **Orbit-wide input-side floors, exhaustively.** The de Groote sandwich factors by side — A-forms depend only on (P, Q), B on (Q, R), C on (R, P) — so three 168×168 GF(2) cost tables cover an entire

orbit (28.4M representatives across the 6 slot variants), with GF(2) side cost a sound lower bound on the $\mathbb{Z}$ side cost (`matmul/orbitscan.py`). For Sun's class: **no representative anywhere in the orbit has input sides below 26 = 13+13** — his representative is side-optimal for the whole class — and *every* representative with sides ≤ 27 (864 per eligible variant, all distinct) was individually re-scored over $\mathbb{Z}$: **none beats 56.** A 55 in Sun's class therefore requires an output side of ≤ 27 on a representative with fat input sides — two below anything observed at this format. The 55 question is now sharply localized.

Everything is exact and replayable: Brent equations over $\mathbb{Z}$ for every scheme, replay-verified straight-line programs, chain re-verification, sign models $\mathbb{Z}$-verified, equivalence witnesses/refutations. Two complete 56-addition programs are committed (`matmul/external/i19-56adds-slp.txt`, appendix; `matmul/external/i12-56adds-slp.txt`).

1 Notation and terminology

- **Sides.** A scheme's additive cost splits into two *input sides* — the A- and B-side forms, over the 9 entries of A and of B — plus one *output side*, the C-side forms computing the 9 outputs from the 23 products. The exact input-side minimizer of §4 (the *side-minimizer* for short) treats the two input sides; the output side is optimized heuristically. Throughout, a bare "sides", a "side floor", and "fat-/slim-sides" refer to the **input** sides; the output side is always named explicitly.
- **Scheme identifiers** (`i12w219c23ci-008`, `i2w201c26fi-000`, …) are the file names of the stored schemes in the public HKS database, used here verbatim as opaque labels; the trailing `-N` indexes files sharing a prefix, and the prefix fields encode HKS's own cataloguing invariants (see the HKS database reference at the end of this note). Nothing in this note relies on their semantics: "class X" always means the de Groote equivalence class of the scheme stored under that name, and every class identity we assert is established by our own exact fingerprint + witness machinery, not by name.
- **de Groote group, class, orbit, representative.** The de Groote symmetry group acts on schemes (sandwiching the three tensors by invertible matrices, plus the S_3 slot symmetries); two schemes are *equivalent* if one is the image of the other. The **orbit** of a scheme is its set of images under the group — identical to its equivalence **class**; a **representative** is any member. An **orbit walk** is a local search over representatives inside one class (moving by small group elements), which can change the additive cost but never the class.
- **sign-SAT** (companion paper, §3.5): the Boolean satisfiability instance whose variables are the signs of the mod-2-supported coefficients. A covering term's sign is the XOR of its three factor signs, and each integer Brent equation with k covering terms and right-hand side δ becomes the cardinality constraint "exactly $(k-\delta)/2$ of the k terms are negative". Its solutions are exactly the valid ± 1 lifts of the mod-2 scheme (worked example in Appendix C).
- **$\mathbb{Z}$-verified**: checked exactly over the integers — all 729 Brent equations evaluated with the scheme's ± 1 coefficients in exact integer arithmetic, zero residuals. (Code output and command comments write plain `Z` for $\mathbb{Z}$.)
- h^*: the star denotes the optimal value — h^* is the smallest helper count h at which the side-minimizer's search succeeds.
- **Closure normal form**: defined in §4 (Method); worked example in Appendix B.

2 Cost model and history

A bilinear scheme with r products computes $M_m = P_m \cdot Q_m$ from linear forms P_m (over the 9 entries of A), Q_m (over B), then each C entry as a linear form over the M_m. The **additive complexity** is the number of binary additions/subtractions in a straight-line program (SLP) computing all forms; unary negation is free; no change of basis (Karstadt–Schwartz accounting is *not* used). This is the model of Mårtensson–Wagner, Stapleton, Perminov, and Sun; Schwartz–Vaknin's 61 lives in the weaker with-basis-change model.

year	count	scheme / method	basis change
1976	98	Laderman, naive form	—
2023	61	Schwartz–Vaknin (pebbling + alt. basis)	**yes**
2024	62	Mårtensson–Wagner, Greedy-Potential on Laderman	no
2025 (Aug)	60	Stapleton (NN + Greedy-Potential); class `i2w201c26fi`	no
this note v1	60 ×3 classes	HKS DB classes + orbit-CSE	no
2025 (Dec 18)	59	MWS, arXiv:2601.05272; class `i19w203c23ci`	no
2025 (Dec 25)	58 (×3)	Perminov, arXiv:2512.21980; classes `i12w219c23ci`, `i46w213c23ci`, `i19w225c4efh`	no
2026 (Apr)	**56 — record**	Sun, arXiv:2604.27645; cyclic image of `i46w213c23ci`	no
this note v2	**56 on two further classes**	`i19w225c4efh`, `i12w219c23ci`; 59 on our three v1 classes (exact input sides + orbit search)	no

3 Why 58 was the pair-extraction wall

Sun's $56 = 13 + 13 + 30$ (A-side + B-side + outputs). Both input sides are *chain-covering*: a single 13-step addition chain per side whose values include all 23 factor rows — every P_m is a bare reference into the chain. Greedy pair extraction builds balanced binary trees of shared pairs; it cannot discover a program in which intermediate *sums of three, four, five inputs* are themselves the reused objects. Measured consequence: our v1 optimizer (and its v2 with kernel and output-reuse moves — both measured, kernels net-harmful, reuse neutral here) plateaus at **58 = 14+14+30 on Sun's own representative**, exactly 1+1 above his sides, with an identical output side. The two missing additions were purely side-structural.

4 The exact input-side minimizer (`matmul/sidemin.py`)

Problem. Given the 23 signed factor rows of one side (vectors in $\{-1, 0, +1\}^9$; rows equal to a basis vector or to ±another row are free), find the minimum number of steps of the form $v = \pm x \pm y$ (x, y earlier values, starting from the 9 basis vectors) such that every distinct multi-term row appears among the values up to global sign. This is Sun's chain-covering structure as a search problem.

Method. Iterative deepening on the number of *helpers* h (chain values that are not target rows); the optimum is (#distinct multi-term targets) $+ h^*$. The search uses what we call the **closure normal form**: because the pool of computed values only ever grows, covering a currently-derivable target can never hurt a later step and always costs exactly one addition — so the search covers every derivable target greedily until none remains (the *closure fixpoint*) and branches **only on which helper value to insert at a fixpoint**. Any optimal chain can be reordered into this form (an exchange argument: coverable-target steps commute forward past helper insertions), so restricting the search to it loses nothing. With one helper left, the helper must directly complete some uncovered target, shrinking the last level to an "enabling set". Helpers may be arbitrary integer vectors, doublings included — strictly more general than the pure/one-aux model of Sun's certificates, so an exhaustion at slack h here is the stronger impossibility statement.

Calibration against Sun's own certificates. His verifier proves each of his sides optimal by a reachability argument (no 12-addition chain covers U's 12 targets; no 12-addition chain with any single auxiliary covers V's 11). Our search independently reproduces all of it — U: 12 targets, $h^* = 1$, 13 additions (3 search nodes); V: 11 targets, $h^* = 2$, 13 additions (11 nodes); $h = 0$ and $h = 1$ exhaustions match his two impossibility certificates — in 0.3 s total, every chain replay-verified. Micro-optima (5 hand-checkable cases) also pass.

Scoring a scheme. Sign models are enumerated by sign-SAT with blocking clauses and $\mathbb{Z}$-verified; per model the two input sides are minimized exactly and the output side is optimized by replay-verified restart-greedy CSE (the v1 pair extractor — on the relevant representatives its counts coincide with the v2 optimizer's, and the output side is where all remaining heuristic slack lives). Side costs were sign-model invariant on every representative tested.

5 Results

Class-by-class (all schemes Brent-verified over $\mathbb{Z}$; class identities by exact fingerprint + witness machinery, cross-checked against the HKS database; "published" = the count in the source paper for that representative):

de Groote class	publ.	this note (A+B+C)	representative
`i46w213c23ci` (= Sun 56,	56	56 = 13+13+30 tied, 24/24 models	Y. Sun's rep
= Perminov cn120 58)		57 = 13+16+28 (C=28 **in class**)	D. Perminov's cn120 rep
`i19w225c4efh` (= cn122)	58	**56 = 13+14+29**	D. Perminov's rep
`i12w219c23ci` (= cn119)	58	57 = 13+15+29	D. Perminov's rep
		56 = 13+14+29	orbit rep (ours)
`i19w203c23ci` (= MWS 59)	59	58 = 14+15+29	E. Mårtensson et al. rep
`i106w191c347g` (ours, v1)	60	**59 = 15+15+29**	v1 orbit rep
`i106w191c23ci` (ours, v1)	60	**59 = 15+15+29**	v1 orbit rep
`i107w189c48ae` (ours, v1)	60	**59 = 15+15+29**	v1 orbit rep
`i2w201c26fi` (Stapleton)	60	60 = 16+16+28	v1 orbit rep

Notes.

- The cn122 result required **no search over representatives at all**: exact side minimization applied to Perminov's own published scheme yields 13+14 sides against pair-greedy's 14+15 — the two additions separating 58 from 56 on that class were entirely side-structural, mirroring the Sun-vs-58 diagnosis.
- The cn119 56-representative was found by a hill-climb over the de Groote orbit scoring exact input sides + greedy-C (`matmul/orbit55.py`, 45 min), then confirmed at high effort (24 sign models × 800 restarts) and confirmed in-class by exact equivalence to cn119 (witness; one class).
- Three classes at 56, pairwise inequivalent (exact refutations among sun56/cn119/cn122 and their images). Before: one.
- Our three v1 classes improve 60 → 59 by exact input sides alone — tying the *previous* (MWS-era) record on classes of our own — and the chain's other counts all drop or tie on their own representatives. Stapleton's class alone resists below 60; notably its representative has the cheapest output side seen anywhere (28) paired with the fattest input sides (16+16).

Attempted and failed at 55 so far (honest negatives, each bounded): four 45-minute orbit climbs (7.2k–8.4k proposals each) from both ends of Sun's class and from cn119/cn122 all converge to 56; 45-minute climbs on the MWS class and one of ours stop at 58 and 59 respectively; deeper sign-model/restart budgets (48 × 800) move nothing; output-reuse moves (the trick behind the 4×4 de-complexification of Dumas–Pernet–Sedoglavic) gain zero on these representatives (28/29/30 unchanged).

6 Orbit-wide input-side floors and the localization of 55

The de Groote sandwich (PAQ^{-1}, QBR^{-1}, $R\tilde{C}P^{-1}$) factors by side: the A-form multiset depends only on (P, Q), B on (Q, R), C on (R, P). So three 168 × 168 tables of *exact GF(2) side costs* (XOR chain covering — same algorithm on 9-bit masks, sub-ms per cell) plus a min-plus scan cover **every representative of an orbit**: $168^3 \approx 4.7$M sandwiches per slot variant, 28.4M per class, in ∼minutes — no hill-climbing, no misses. A GF(2) side cost lower-bounds the $\mathbb{Z}$ side cost of the same representative (reduce any $\mathbb{Z}$-chain mod 2), so the scan yields sound orbit-wide side floors (`matmul/orbitscan.py`).

Sun's class (`i46w213c23ci`), all six variants:

variant	GF(2) sides floor	candidates ≤ 57 (est)
0	**26**	432
1	28	216
2	29	324
3	**26**	444
4	29	252
5	28	216

No representative of the class, in any variant, has input sides below $26 = 13+13$: **Sun's representative is side-optimal for his entire equivalence class.** Furthermore every representative with sides ≤ 27 (the only ones that could give 55 with an output side of 28–29) was enumerated — 864 distinct schemes in each of the two floor-26 variants — and each was individually re-scored over $\mathbb{Z}$ at high effort: **best = 56 = 13+13+30, no exceptions.** Combining:

- a 55 with sides 26–27 in Sun's class is excluded up to the output heuristic's reliability on those 1,728 exhaustive re-scores;
- a 55 with sides 28–29 requires an output side of 26–27 — at least two below the best output count ever observed at this format (28, seen in exactly two classes), on representatives whose C tensors we scored throughout the scans.

The same machinery applied to `i19w225c4efh` (cn122) shows side floors of 27–28, so its $56 = 13+14+29$ sits one addition above its own side floor; a 55 there needs sides $27 + C$ 28. These are now *concrete, finite* search targets rather than open-ended hunts.

7 The database-wide input-side-floor census (v3 addition, 2026-07-05)

With the table machinery ported to Rust (`src/floors.rs`, $\sim$300×; a sound nt-prefilter for sweep mode) and exact Z-rescoring in-process (`src/zrescore.rs`, $\sim$150×), the orbit-side analysis was run over the **entire HKS database — all 17,376 classes, every representative of every orbit** ($\sim 10^{11}$ representatives covered by table decomposition). Screened side-floor histogram: **26: 4 classes; 27: 526; 28: 12,159; 29: 4,326; 30: 357; 31: 4.**

- **Floor 26 (Sun-grade sides) exists in exactly four classes**: Sun's `i46w213c23ci`, its family siblings `i46w205c23ci` and `i46w221c23ci`, and `i73w191c236f` (floors exact-confirmed unscreened). Exhausting every sides ≤ 27 representative of the three new ones (4,320 reps, exact $\mathbb{Z}$-rescoring): best totals **57, 57, 58** — chain-coverable sides are *not* sufficient; Sun's class is the only one in the published record that pairs them with an output side ≤ 30. (Control: the same run reproduces Sun's class at 56 from its 1,728 slim-sides reps.)
- **Of the 526 screened floor-27 classes, 80 truly admit sides 27** (the rest are prefilter lower-bound artifacts, filtered exactly as designed). Exhausting all of them: best totals 56 ×2, 57 ×6, 58 ×27, 59 ×26, 60 ×18, 61 ×1 — and the two 56s are precisely `i12w219c23ci` and `i19w225c4efh`: **the blind database sweep independently rediscovers both classes of §5**, with no input from this note's earlier results.
- Consequently, over the entire published database: **exactly three classes attain 56; no slim-sides representative of any class attains 55 anywhere**; and any 55 at rank 23 within known classes must be a fat-sides representative (sides ≥ 28) with an output side of ≤ 27 — two additions below the best output count ever observed at this format. The per-class closure tool for that last window is the exact XOR-SLP (an SLP over GF(2)) lower-bound program of §11 (`matmul/cxlb.py`: UNSAT at $k \Rightarrow C_{\mathbb{Z}} \geq k+1$, DRAT-certifiable); its calibration brackets the two key cells at GF(2) C-min $\in \{29, 30\}$ (Sun's rep) and $\in \{27, 28\}$ (the C=28 rep of Sun's class).

Committed artifacts: `matmul/db_floor_census.csv` (17,376 rows), `matmul/f27_campaign_results.csv` (80 rows), campaign driver `matmul/found55/f27_campaign.sh`.

8 The 60-era results (v1 of this note, compressed)

Kept because the discovery path and the negative are still informative; full text in git history.

- **Full-database CSE screen** (all 17,376 HKS schemes; apparently the first such sweep): exactly three classes reach 61 from their stored representatives; the classics: Laderman 62, Smirnov 68 (sparsest $\neq$ most CSE-able). Committed: `matmul/cse_screen_top200.csv`.

- **Orbit search over representatives** (transvection moves): the optimized count is **not** a de Groote invariant — within Stapleton's class the stored representative plateaus at 62 while his reaches 60; our three v1 classes went 61 → 60 the same way. The representative axis has now been worth ≥ 2 additions twice over: 62 → 60 (v1) and 58 → 56 (Sun vs Perminov, one group element apart — classified here).
- **Reclassification of Stapleton's scheme**: de Groote-equivalent to HKS `i2w201c26fi-000` (exact witness) — his neural pipeline rediscovered a 2019 database class; no such check appears in his note. Perminov's schemes are database classes too (IDs above), as is Sun's; one of Perminov's non-record schemes (`naive88`) even lands in one of *our* v1 classes (`i107w189c48ae`).
- **The greedy-family ceiling**: >20,000 well-mixed orbit proposals per class × 16 sign models × 128 restarts never moved any of the four 60-classes below 60. We wrote, honestly, that this was "either a true barrier at 60 or a shared ceiling of the greedy-CSE family". The chain — and now our own exact input-side results — settle it: it was the family ceiling, and the missing structure was chain-covering input sides. The v1 measurement stands as a clean characterization of what pair-extraction alone can and cannot see.

9 What is new here, plainly

- The input-side subproblem of additive complexity solved *exactly*, with optimality certificates cross-validated against Sun's, and the observation that this alone — no new schemes, no orbit search — turns Perminov's published cn122 into a record-tying 56.
- **Two new 56-addition classes** (`i19w225c4efh`, `i12w219c23ci`), tripling the number of known record-count classes; complete replay-verified 56-addition programs committed for both.
- Exhaustive orbit-wide side floors (a sound lower-bound method over 28.4M representatives per class in minutes), the database-wide census over all 17,376 classes, and, for Sun's class, side-optimality of his representative over the whole class plus an exhaustive $\mathbb{Z}$-re-score of every slim-sides representative — the sharpest localization yet of where a 55 could live.
- Improved counts on every other class in the record chain's history on their own representatives (59 → 58, 58 → 57, 60 → 59 ×3).

10 Reproduction

```
cd matmul
# exact side-minimizer: micro-optima + Sun's U/V optima + his
# impossibility certificates (0.3 s)
python3 sidemin.py selftest

# the record chain, exact input sides + greedy C (Z-verified sign models):
# sun56 -> 56 = 13+13+30 on all models; cn122 -> 56 = 13+14+29;
# cn119 -> 57; cn120 -> 57 = 13+16+28; mws59 -> 58
python3 sidemin.py --models 24 perminov_cache/bits/sun56.bits \
    perminov_cache/bits/cr58-cn122.bits perminov_cache/bits/cr58-cn119.bits \
    perminov_cache/bits/cr58-cn120.bits perminov_cache/bits/mws59.bits

# the two new 56-classes at high effort, with full SLP emission
python3 sidemin.py --models 24 --c-restarts 800 \
    --emit /tmp/i19.slp external/i19-perminov56.bits
python3 sidemin.py --models 24 --c-restarts 800 \
    --emit /tmp/i12.slp external/i12-orbit56.bits

# class identities: three pairwise-inequivalent 56-classes;
# cn119-orbit rep stays in cn119's class
python3 equiv.py classes perminov_cache/bits/sun56.bits \
    external/i19-perminov56.bits external/i12-orbit56.bits   # expect 3
python3 equiv.py classes external/i12-orbit56.bits \
    perminov_cache/bits/cr58-cn119.bits                     # expect 1
```

```
# our three v1 classes at 59; Stapleton at 60
python3 sidemin.py --models 24 external/i106-orbitbest.bits \
    external/i106b-orbitbest.bits external/i107-orbitbest.bits \
    external/stapleton60-orbitbest.bits

# orbit-wide side floors + candidate re-scoring (Sun's class, ~20 min)
python3 orbitscan.py perminov_cache/bits/sun56.bits --cutoff 57
# exhaustive slim-sides re-score (variants 0+3, ~1 h)
python3 orbitscan.py perminov_cache/bits/sun56.bits \
    --exhaust-sides 27 --variants 03 --models 8 --crestarts 300

# orbit hill-climb used for the cn119 56-representative (stochastic)
python3 orbit55.py perminov_cache/bits/cr58-cn119.bits --minutes 45 \
    --out /tmp/cn119-best.bits

# HKS class IDs of every chain scheme (needs the DB cache;
# dbcheck.py fetch convert first if absent)
python3 perminov_cache/q2_check.py
```

Deterministic caveats: kissat may return different sign models across versions (each returned model is $\mathbb{Z}$-verified); orbit climbs are stochastic — the *committed representatives* make every headline count deterministic to check.

11 Limitations and the route to 55

Input sides are now exact; **the output side is the open flank** — all counts above use replay-verified greedy CSE there, and nothing bounds C from below beyond trivialities. The observed C landscape at this format: 28 in exactly two classes (Stapleton's, and Perminov's cn120 representative of Sun's class), 29–30 elsewhere; reuse moves gain nothing. The concrete open combinations:

- **C-side exact minimization or lower bounds.** The side-minimizer does not transfer (the C side has $\sim$20 helpers' worth of slack); candidate tools are transposition-principle arguments, SAT/ILP with structure, or a chain-covering generalization to fan-in trees. Any $C \leq 27$ on a fat-sides representative of Sun's class, or $C \leq 28$ on a sides-27 representative of cn122's class, is an outright 55. First steps exist: `matmul/cxlb.py` decides "k XOR additions suffice" exactly (UNSAT $\Rightarrow$ sound $\mathbb{Z}$ bound, DRAT-certifiable); its boundary instances resist all current solvers at 10–30 minute budgets and are being submitted as SAT-competition benchmarks (`doc/matmul_cxlb_satcomp.pdf`).
- **Class sweep: done** (§7) — the census found no class with a floor below 26 and no new 56-class; within the published record, the fat-sides window is the only route left.
- The r=22 question (challenge 4) remains open and is where any count would change complexity theory rather than constants.

References

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HKS scheme database (the 17,376-scheme corpus this note classifies against; source of the scheme identifiers): http://www.algebra.uni-linz.ac.at/research/matrix-multiplication/ — the `www.` host is required (the bare domain does not resolve; the site serves plain http, with a self-signed cert on https). There is no separate GitHub mirror of the database; the related GitHub artifact is the authors' SAT-benchmark repo https://github.com/marijnheule/matrix-challenges. Our reproduction does not depend on the site being reachable: the corpus is the snapshot `schemes.tgz` (43,134,227 bytes, Last-Modified 2020-08-07, sha256 `4bc8132644504a917e3c076f64df8e6619fb67c55670179853ae5fdb1583074f`), fetched once and cached; all class results are pinned to it. A permanent, content-verified archival mirror of that exact snapshot is deposited at https://doi.org/10.5281/zenodo.21209925 (cite the HKS papers for the schemes, not the mirror).

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Appendix A: a complete 56-addition program (class `i19w225c4efh`)

Exact input sides (13 + 14, both provably minimal for this representative by the side-minimizer's exhaustion), greedy output side (29). On disk: `matmul/external/i19-56adds-slp.txt` with bits at `matmul/external/i19-perminov56.bits` (= Perminov's published cn122 scheme, our sign model m0). Note the A-side: one long addition chain — the structure pair extraction cannot express.

```
## A-side: 13 additions
aw0 = a13 + a23          P1 = aw2     P13 = a23
aw1 = a11 + aw0          P2 = aw8     P14 = aw7
aw2 = a12 + aw1          P3 = a11     P15 = a21
aw3 = -a23 + aw2         P4 = a32     P16 = a31
aw4 = a22 + aw2          P5 = a11     P17 = a31
aw5 = a33 + aw3          P6 = aw10    P18 = aw0
aw6 = a32 + aw5          P7 = aw6     P19 = aw5
aw7 = a21 + aw4          P8 = a33     P20 = aw12
aw8 = a31 + aw7          P9 = aw9     P21 = a12
aw9 = a32 + aw8          P10 = aw1    P22 = a22
aw10 = -a12 + aw5        P11 = a33    P23 = aw4
aw11 = a21 + a31         P12 = aw3
aw12 = aw1 + aw11

## B-side: 14 additions
bw0 = b13 - b33          Q1 = bw10    Q13 = b32
bw1 = b11 - b21          Q2 = bw6     Q14 = bw2
bw2 = b11 + b13          Q3 = bw0     Q15 = b12
bw3 = b11 - b31          Q4 = b22     Q16 = b12
bw4 = b23 - b33          Q5 = bw9     Q17 = b13
bw5 = bw1 - bw4          Q6 = bw4     Q18 = bw3
bw6 = b13 + bw1          Q7 = b23     Q19 = bw12
bw7 = -b23 + bw6         Q8 = b31     Q20 = bw1
bw8 = b32 + bw3          Q9 = b21     Q21 = bw11
bw9 = -b12 + bw8         Q10 = bw5    Q22 = b22
bw10 = -bw5 + bw8        Q11 = b32    Q23 = bw7
bw11 = -b22 + bw10       Q12 = bw13
bw12 = -b21 + bw10
bw13 = b23 - bw12

## outputs: 29 additions
cw0 = M11 -M19           C11 = cw4 -cw6
cw1 = M8 -cw0            C12 = -M21 -M5 +M10 -cw6
cw2 = M6 -cw1            C13 = M12 +M3 -cw2
cw3 = M1 -M13            C21 = -cw3 +cw7 +cw8
cw4 = M10 +cw2           C22 = M13 +M15 +M22
cw5 = M2 -M17            C23 = -M12 -M23 +cw5 -cw8
cw6 = M18 -cw3           C31 = M8 +M9 -cw7
cw7 = M14 -cw5           C32 = M11 +M16 +M4
cw8 = M20 -cw4           C33 = -M12 +M17 +M7 -cw1
```

Appendix B: the side-minimizer on a worked example

Rows (targets) over base variables a, b, c, d: $T_1 = a + b$ and $T_2 = a + b + c + d$. Both have ≥ 2 terms, are distinct up to global sign, and neither is a basis vector, so $\mathrm{nt} = 2$ and any chain needs at least 2 additions.

Slack $h = 0$ (no helpers — every addition must create a target): the pool starts as $\{a, b, c, d\}$. $T_1 = a + b$ is derivable (both operands in the pool): cover it; the pool grows to $\{a, b, c, d, a{+}b\}$. Now T_2 must be $x \pm y$ with x, y in the pool: subtracting each pool value from T_2 gives $c{+}d$, $a{+}c{+}d$, $b{+}c{+}d$, $a{+}b{+}c$, $a{+}b{+}d$ — none in the pool, so the closure is at a fixpoint with T_2 uncovered, and $h = 0$ is **exhausted: 2 additions are impossible.**

Slack $h = 1$: at that same fixpoint the *enabling set* — values u that would directly complete an uncovered target as $T_2 = x \pm u$ — is $\{c{+}d,\ a{+}c{+}d,\ b{+}c{+}d,\ a{+}b{+}c,\ a{+}b{+}d, \dots\}$; of these, $u = c + d$ is itself derivable from the pool (c and d are basis vectors). Insert it as the helper, and the closure resumes: $T_2 = (a{+}b) + (c{+}d)$.

The optimum is $\mathrm{nt} + h^* = 2 + 1 =$ **3 additions**, found in 3 search nodes; the emitted chain is

```
w0 = a + b        (covers T1)
w1 = c + d        (helper)
w2 = w0 + w1      (covers T2)
```

replay-verified by expansion. Global signs are free throughout: a row $-a - b$ is the same target as $a + b$ (a chain value may be used negated at no cost).

Appendix C: $\mathbb{Z}$-scoring a scheme, end to end (sun56)

The pipeline of §4 on the record representative, with every number below reproduced by `sidemin.py --models 24 perminov_cache/bits/sun56.bits`:

1. **Mod-2 gate.** The 621-bit vector satisfies all 729 Brent equations over GF(2).
2. **Sign-SAT.** The support has 175 nonzero coefficients (175 sign variables); the 729 integer Brent equations have 453 covering terms in total, and the instance has 4,269 clauses. Two concrete equations: one type-3 equation is covered by $k = 3$ terms (the products M_2, M_{17}, M_{23}) with right-hand side 1, so **exactly** $(3-1)/2 = 1$ **of the three terms must be negative**; a neighboring rhs-0 equation is covered by $k = 4$ terms (M_2, M_{17}, M_{20}, M_{23}), so exactly 2 of 4 are negative. kissat solves the instance in milliseconds; each returned model is $\mathbb{Z}$-verified (all 729 equations, exact integer arithmetic) before use.
3. **Exact input sides** (model m0). The 23 A-rows contain 12 distinct multi-term targets; $h = 0$ is exhausted and $h^* = 1$ succeeds: A-side $= 12 + 1 =$ **13** additions (3 search nodes). The B-rows contain 11 distinct targets; $h = 0$ and $h = 1$ are exhausted, $h^* = 2$ succeeds: B-side $=$ **13** (11 nodes). These reproduce Sun's own input-side optima and impossibility certificates.
4. **Greedy output side.** Deterministic pair extraction gives 31 additions — its first extraction is $w_0 = M_{13} - M_{23}$, a signed pair shared by 3 of the 9 output forms. Randomized tie-breaking restarts improve this: best over 300 restarts $=$ **30** (the value used throughout; every emitted program is replay-verified).
5. **Total.** $13 + 13 + 30 =$ **56**, on every one of the 24 enumerated sign models.